

1) eigen vector

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eigen value

$$1. \quad Au = \lambda u, u \neq 0 \quad \xrightarrow[\text{transpose}]{\text{conjugate}} \quad (Au)^* = (\lambda u)^*$$

$$\longrightarrow u^* A^* = \lambda^* u^* \quad \xrightarrow[A \text{ is symmetric}]{A \text{ is real}} \quad u^* A = \lambda^* u^*$$

$$\longrightarrow u^* A u = \lambda^* u^* u \quad (\text{I})$$

$$Au = \lambda u \longrightarrow u^* A u = u^* \lambda u \quad (\text{I}) \longrightarrow \lambda = \lambda^*, \lambda \text{ is real} \checkmark$$

$$A \text{ is PD} \longrightarrow \forall x \in \mathbb{R}^n: x^T A x > 0$$

$$\text{we put } x = u \Rightarrow u^T (Au) = u^T \lambda u = \lambda u^T u \quad (\text{II})$$

$$u^T u > 0 \quad (\text{II}) \longrightarrow \lambda > 0 \checkmark$$

2.

$$a) \quad \forall x \in \mathbb{R}^n \Rightarrow f(x) \geq 0, \quad b) \quad f(x) = 0 \text{ iff } x = 0$$

$$c) \quad f(tx) = |t| f(x), \quad d) \quad f(x+y) \leq f(x) + f(y)$$

$$a, b: \quad A \text{ is PD} \longrightarrow x^T A x > 0, \quad x^T A x = 0 \text{ iff } x = 0 \checkmark$$

$$c: \quad f(tx) = \sqrt{(tx)^T A (tx)} = \sqrt{t^2 x^T A x} = |t| \sqrt{x^T A x} \checkmark$$

d: we define the following inner product:

$$\langle x, y \rangle_A = x^T A y, \quad \text{Cauchy Schwarz:}$$

$$|\langle x, y \rangle_A| \leq \sqrt{|\langle x, x \rangle_A|} \sqrt{|\langle y, y \rangle_A|} = f(x) f(y) \quad (\text{I})$$

$$f(u+y)^2 = \langle u+y, u+y \rangle_A = f(u)^2 + f(y)^2 + 2\langle u, y \rangle_A$$

$$\textcircled{I}, \quad f(u+y)^2 \leq f(u)^2 + f(y)^2 + 2f(u)f(y) = (f(u)+f(y))^2$$

$$\sqrt{\quad}, \quad f(u+y) \leq f(u)+f(y) \quad \checkmark$$

$$3. \quad AA^T = I \xrightarrow{T} (A^{-1})^T A^T = I \xrightarrow{A=A^T} (A^{-1})^T A = I$$

$$\rightarrow (A^{-1})^T = A^{-1} \rightarrow A^{-1} \text{ is symmetric } \checkmark$$

$$\text{we define } y = A^{-1}u, \text{ we know } u^T A u > 0 \rightarrow$$

$$\rightarrow u^T A^{-1} u = (Ay)^T A^{-1} (Ay) = y^T A y > 0 \rightarrow A^{-1} \text{ is PD } \checkmark$$

$$A^{-1} = (Q \Lambda Q^T)^{-1} = Q \Lambda^{-1} Q^T$$

$$\Lambda = \text{diag}(\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}) \rightarrow \text{eigenvalues of } A = \frac{1}{\lambda_1} \dots \frac{1}{\lambda_n} \quad \checkmark$$

2)

$$1. \quad XY - YX = I \xrightarrow{\text{tr}} \text{tr}(XY - YX) = \text{tr}(I)$$

$$\rightarrow \text{tr}(XY) - \text{tr}(YX) = n, \quad \text{tr}(AB) = \text{tr}(BA)$$

$$\rightarrow \text{tr}(XY) = \text{tr}(YX) \rightarrow 0 = n \quad \times$$

so no matrix can be found

$$2. \|Aq\|_2 = (Aq)^T (Aq) = q^T \underbrace{A^T A}_I q = q^T q = \|q\|_2$$

$$3) 1. \text{ if } df = \text{tr}(G^T dx) \rightarrow \nabla_x f(x) = G$$

$$f(x) = \text{tr}(x^T A x B) \quad \rightsquigarrow \text{linear operators exchanged}$$

$$\rightarrow df = d[\text{tr}(x^T A x B)] = \text{tr}[d(x^T A x B)]$$

$$= \text{tr}[dx^T A x B + x^T A dx B]$$

$$= \text{tr}[dx^T A x B] + \text{tr}[x^T A dx B]$$

$$\text{tr}[dx^T A x B] = \text{tr}[(dx^T A x B)^T] = \text{tr}[B^T x^T A^T dx]$$

$$\text{tr}[x^T A dx B] = \text{tr}[B x^T A dx]$$

$$\rightarrow df = \text{tr}(B^T A^T A dx) + \text{tr}(B x^T A dx)$$

$$= \text{tr}(\underbrace{(B^T A^T A + B A^T A)}_M dx)$$

$$\rightarrow G = \nabla_x f(x) = M^T = A x B + A^T x B^T$$

$$A = A^T, B = B^T \rightarrow \nabla_x f(x) = 2 A x B$$

$$2. L(x, \Lambda) = f(x) - \text{tr}(\Lambda^T H) \quad \begin{matrix} \nearrow \text{constrain: } H=0 \\ \searrow \text{lagrange multiplier} \end{matrix}$$

$$\rightarrow L(x, \Lambda) = \text{tr}(x^T A x B) - \text{tr}(\Lambda(x^T x - I))$$

$$3. \quad S = \sqrt{X}^T X \sqrt{X} = \int_{1 \times n} X_{n \times n} \sqrt{X_1} = \text{some scalar}$$

$$\rightarrow S = S^T \rightarrow S^T = (\sqrt{X}^T X \sqrt{X})^T = \sqrt{X}^T X^T \sqrt{X}$$

$$X^T = -X \rightarrow S^T = -\sqrt{X}^T X \sqrt{X} = -S$$

$$\rightarrow S = -S \rightarrow S = 0 \rightarrow \sqrt{X}^T X \sqrt{X} = 0$$

$$L() \quad f(a_1, a_2) = (a - a_1)^2 + b(a_2 - a_1^2)^2$$

$$1. \quad \nabla f(a) = \begin{bmatrix} \frac{\partial f}{\partial a_1} \\ \frac{\partial f}{\partial a_2} \end{bmatrix} = \begin{bmatrix} -2(a - a_1) - 4b a_1 (a_2 - a_1^2) \\ 2b(a_2 - a_1^2) \end{bmatrix}$$

$$H(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial a_1^2} & \frac{\partial^2 f}{\partial a_1 \partial a_2} \\ \frac{\partial^2 f}{\partial a_1 \partial a_2} & \frac{\partial^2 f}{\partial a_2^2} \end{bmatrix} = \begin{bmatrix} 2 - 4b a_2 + 12b a_1^2 & -4b a_1 \\ -4b a_1 & 2b \end{bmatrix}$$

$$2. \quad \nabla f(a) = 0 \rightarrow \begin{cases} \frac{\partial f}{\partial a_1} = 0 \rightarrow 2b(a_2 - a_1^2) = 0 \rightarrow a_2 = a_1^2 \\ \frac{\partial f}{\partial a_2} = 0 \rightarrow -2(a - a_1) - 4b a_2 = 0 \rightarrow a_1 = a \end{cases}$$

$$\rightarrow X^* = (a, a^2), \quad f \geq 0 \text{ everywhere} \rightarrow (a, a^2)$$

is global minimum

checking with H: H must be PD, so by leading principal minors:

$$H_{11} = 2 + 8b a^2 > 0 \quad \checkmark$$

$$\det(H) = 4b > 0 \quad \checkmark \Rightarrow H \text{ is PD}$$

3. GR: $\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$, newton: $\theta_{t+1} = \theta_t - H_t^{-1} \nabla f(\theta_t)$

the convergence of both methods is governed by hessian

we define: $\frac{\lambda_{\max}(H^*)}{\lambda_{\min}(H^*)} = K$ as the condition number

$\Rightarrow \text{tr}(H^*) = 2 + 8ba^2 + 2b$, $\det(H^*) = 4b$

$\rightarrow K \approx 21a^2 + 16ba^4$, which would make a very ill-conditioned Hessian for large a, b .

local linear convergence rate:

$\|x_{t+1} - x^*\| \leq \left(\frac{K-1}{K+1}\right) \|x_t - x^*\|$ $\rightarrow$ very close to 1

$\rightarrow$ slow convergence ✓

newton: $\|x_{t+1} - x^*\| \leq C \|x_t - x^*\|^2$ for some $C \geq 0$

$\rightarrow$ the convergence is quadratic ✓

we can say that in this case, f has a very narrow

curved parabolic valley, the GR moves $\perp$ to

the current contour, so it forms a zigzag progress,

while newton wraps space by H^{-1} , turning ellipses

into circles. in a round landscape, both methods

work fine.

5)

$$1. \quad f(x_1, x_2) = x_1^2 + x_2^2$$

$$\text{constraints: } g_1 = g = x_1 + x_2 - 2 \leq 0$$

$$g_2 = h = x_1^2 - x_2 \leq 0$$

$$L(\theta, \alpha) = f(\theta) + \sum_i \alpha_i \underbrace{g_i(\theta)}_{\text{multipliers}}$$

$$\rightarrow L(\theta, \alpha) = L(x_1, x_2, \alpha_1, \alpha_2) = x_1^2 + x_2^2 + \alpha_1(x_1 + x_2 - 2) + \alpha_2(x_1^2 - x_2)$$

2. KKT conditions:

$$a. \quad \nabla L(\theta) + \sum_i \alpha_i \nabla g_i(\theta) = 0 \quad (\text{stationary condition})$$

$$\nabla f = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}, \quad \nabla g_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla g_2 = \begin{bmatrix} 2x_1 \\ -1 \end{bmatrix}$$

$$\rightarrow \begin{cases} 2x_1(1 + \alpha_1) + \alpha_1 = 0 & S1 \\ 2x_2 + \alpha_1 - \alpha_2 = 0 & S2 \end{cases} \quad \begin{matrix} \uparrow \\ \text{condition names} \end{matrix}$$

$$b. \quad \begin{cases} g_1 = g = x_1 + x_2 - 2 \leq 0 \\ g_2 = h = x_1^2 - x_2 \leq 0 \end{cases} \quad \begin{matrix} F1 \\ F2 \end{matrix} \quad \begin{matrix} \uparrow \\ \text{feasibility condition} \end{matrix}$$

$$c. \quad \alpha_i \geq 0 \rightarrow \begin{cases} \alpha_1 \geq 0 \\ \alpha_2 \geq 0 \end{cases} \quad \begin{matrix} D1 \\ D2 \end{matrix} \quad \begin{matrix} \uparrow \\ \text{dual feasibility} \end{matrix}$$

$$d. \quad \alpha_1 g_1 = 0, \alpha_2 g_2 = 0 \rightarrow \begin{cases} \alpha_1(x_1 + x_2 - 2) = 0 & "c1" \\ \alpha_2(x_1^2 - x_2) = 0 & "c2" \end{cases}$$

complementary condition

3. we start with complementary slackness

from c_1, c_2 conditions:

case	u_1	u_2
A	0	0
B	>0	0
C	0	>0
D	>0	>0

for case A: $s_1 \rightarrow 2u_1 = 0 \rightarrow u_1 = 0$
 $s_2 \rightarrow 2u_2 = 0 \rightarrow u_2 = 0$

$g_1(0,0) \leq 0, g_2(0,0) \leq 0$ feasibility ✓

$\rightarrow (0,0)$ fits the KKT ✓

B: $u_1 + u_2 - 2 = 0 \rightarrow u_2 = 2 - u_1$

$s_1 \rightarrow 2u_1 + u_1 = 0$

$s_2 \rightarrow 2u_2 + u_1 = 0$

$\left. \begin{array}{l} u_2 = 2 - u_1 \\ 2u_2 + u_1 = 0 \end{array} \right\} \rightarrow u_1 = u_2 = 1$

but $u_1 = -2 < 0 \rightarrow$ condition C, fails X

the same approach in part A, B can show that

cases C, D also don't offer any KKT point

$\rightarrow x^* = (0,0), f(x^*) = 0, u_1 = u_2 = 0$

sufficiency theorem for convex problems:

if f , all g_i are convex function, all h_j are affine $\rightarrow$ KKT conditions are sufficient for global optimality

$$f(x_1, x_2) = x_1^2 + x_2^2 \rightarrow H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \text{ PD } \checkmark \text{ convex}$$

$$g_1 = x_1 + x_2 - 2 \text{ is affine} \rightarrow \text{convex } \checkmark$$

$$g_2 = x_1^2 - x_2^2 \rightarrow H = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \rightarrow \text{PSD} \rightarrow \text{convex } \checkmark$$

$\rightarrow (0, 0)$ is a global minimizer

6)

$$1. \quad J(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(\theta; x_i, y_i) + \frac{\lambda}{2} \|\theta\|_2^2$$

$$\xrightarrow{H} \nabla^2 J(\theta) = \frac{1}{N} \sum_{i=1}^N \nabla^2 \ell(\theta, x_i, y_i) + \frac{\lambda}{2} \nabla^2 (\|\theta\|_2^2)$$

$$\|\theta\|_2^2 = [\theta_1^2 \dots \theta_n^2]^T \rightarrow \nabla \|\theta\|_2^2 = [2\theta_1 \dots 2\theta_n]^T$$

$$\rightarrow \nabla^2 (\|\theta\|_2^2) = 2I$$

$$\rightarrow \nabla^2 J(\theta) = \frac{1}{N} \sum_{i=1}^N \nabla^2 \ell(\theta; x_i, y_i) + \lambda I$$

all L_i are convex

$$\rightarrow \nabla^2 L_i(\theta) \succeq 0 \quad (\text{PSD})$$

$$\rightarrow \nabla^2 J = \sum_{i=1}^n \nabla^2 L_i + \lambda I \succeq 0 \quad (\text{PD}) \quad \checkmark$$

$\rightarrow J(\theta)$ is strictly convex, that matters

because it means:

the global optimum is unique, every local min is global.

2. $\theta_{t+1} = \theta_t + \eta(-g_t)$, g_t was calculated in part 1

$$g_t = \nabla_{\theta} L(\theta; x_i, y_i) + \lambda \theta_t$$

$$\rightarrow \theta_{t+1} = \theta_t - \eta (\nabla_{\theta} L(\theta; x_i, y_i) + \lambda \theta_t)$$

3. SGD uses noisy gradient $g_t = \nabla J(\theta_t) + \text{noise}$

$$\mathbb{E}[g_t] = \nabla J(\theta_t) \rightarrow \text{SGD moves in correct descent}$$

η_t must gradually decrease, typically conditions can be

$$\sum_t \eta_t = \infty, \quad \sum_t \eta_t^2 < \infty, \quad \eta_t > 0, \quad \text{for convex function:}$$
$$J(\theta) - J(\theta^*) \geq \nabla g(\theta)^T (\theta - \theta^*) \rightarrow \text{SGD moves to}$$

global min in average, strict convexity says θ^* is unique.

$$\bullet \quad A \rightarrow A^*$$

7) first we make sure the problem is convex

$$f(\mathbf{a}) = -\sum_{i=1}^n \log(\alpha_i + a_i)$$

$$\rightarrow H_f = \nabla^2 f(\mathbf{a}) = \text{diag}\left(\frac{1}{(\alpha_1 + a_1)^2}, \dots, \frac{1}{(\alpha_n + a_n)^2}\right) \succ 0 \quad (\text{PD})$$

$\rightarrow f$ is strictly convex

$g_i = -a_i \leq 0$, $h(\mathbf{a}) = \mathbf{1}^T \mathbf{a} - 1$ are affine hence convex

$\rightarrow$ Problem is strictly convex $\rightarrow$ unique solution

now we define the Lagrangian:

$$L = f(\mathbf{a}) + \sum_i u_i g_i(\mathbf{a}) + \lambda h(\mathbf{a})$$

$$= f(\mathbf{a}) - \sum_{i=1}^n u_i a_i + \lambda \sum_{i=1}^n (a_i - 1)$$

KKT conditions:

a. Stationary

$$\nabla L = 0 \rightarrow \frac{\partial L}{\partial a_i} = 0 \quad \forall i$$

$$\rightarrow -\frac{1}{\alpha_i + a_i} - u_i + \lambda = 0$$

feasibility: $x_i \geq 0 \quad \forall i$, $\sum_{i=1}^n a_i = 1$

dual feasibility: $u_i \geq 0 \quad \forall i$

slackness: $u_i g_i = 0 \quad \forall i$

two cases from slackness:

$$A: \alpha_i > 0, u_i = 0 \xrightarrow{\text{stationary condition}} \alpha_i = \frac{1}{\lambda} - \alpha_i$$

$$\text{feasibility} \rightarrow \alpha_i = \frac{1}{\lambda} - \alpha_i \geq 0 \rightarrow \frac{1}{\lambda} \geq \alpha_i$$

$$B: \alpha_i = 0, u_i > 0 \xrightarrow{s} u_i = \lambda - \frac{1}{\alpha_i} \rightarrow \lambda \geq \frac{1}{\alpha_i}$$

$$\Rightarrow x_i^* = \max \left\{ \frac{1}{\lambda} - \alpha_i, 0 \right\}$$

$$\text{if we define: } \lambda \triangleq \frac{1}{\lambda}$$

$$\rightarrow x_i^* = \max \{ \lambda - \alpha_i, 0 \} \text{ which means if}$$

$\lambda < \alpha_i$ channel i gets no power

$$\sum_{i=1}^n \alpha_i^* = 1 \rightarrow \sum_{i \in A} (\lambda - \alpha_i) = 1$$

$$\text{where } A = \{ i; \lambda > \alpha_i \}$$

$$\rightarrow |A| \cdot \lambda - \sum_{i \in A} \alpha_i = 1 \Rightarrow \lambda = \frac{1 + \sum_{i \in A} \alpha_i}{|A|}$$

$$x_i^* = \max \{ \lambda - \alpha_i, 0 \} \text{ where}$$

8)

1. by Rayleigh quotient, we know that:

$$\lambda_1(M) = \max_{\|q\|=1} q^T M q \quad \text{for some matrix } M$$

$$\rightarrow \lambda_1(uv) = 2^T (uv) 2 = 2^T u (v^T 2) \quad \text{for some } \|v\|=1$$

we define following inner product and then apply Cauchy Schwarz:

$$\langle x, y \rangle = x^T u y$$

$$\rightarrow \|x^T u y\| \leq \sqrt{x^T u x} \sqrt{y^T u y}$$

$$x=2, y=v \rightarrow 2^T u (v^T 2) \leq \sqrt{2^T u 2} \sqrt{(v^T 2)^T u (v^T 2)} \quad \textcircled{I}$$

$$\text{by Rayleigh} \rightarrow (v^T 2)^T u (v^T 2) \leq \lambda_1(u) \|v^T 2\|^2, \|v\|=1$$

$$= 2^T v^T 2 = 2^T v^T 2 =$$

(v = v^T)

$$\rightarrow (v^T 2)^T u (v^T 2) \leq \lambda_1(u) \lambda_1(v)^2 \quad \textcircled{II}$$

$$\textcircled{II} \textcircled{I} \rightarrow \lambda_1(uv) \leq \sqrt{\lambda_1(u)} \cdot \sqrt{\lambda_1(u) \lambda_1(v)^2} = \lambda_1(u) \lambda_1(v)$$

✓

$$2. \lambda_1(c_K(M)) = \prod_{i=1}^K \lambda_i(M) \quad (\text{I})$$

where $c_K(M)$ is the K -th compound matrix of M

$$\rightarrow c_K(uv) = c_K(u) c_K(v)$$

we know that $u \succeq 0 \implies c_K(u) \succeq 0$

$$\textcircled{\text{I}} \rightarrow \lambda_1(c_K(u) c_K(v)) \leq \lambda_1(c_K(u)) \lambda_1(c_K(v))$$

$$\rightarrow \prod_{i=1}^K \lambda_i(uv) \leq \prod_{i=1}^K \lambda_i(u) \prod_{i=1}^K \lambda_i(v) \quad \checkmark$$

3. by SVD decomposition:

$$A = U_A \Sigma_A^T, \quad B = U_B \Sigma_B^T$$

$$\text{Tr}(A^T B) = \text{tr}(\Sigma_A^T U_A^T U_B \Sigma_B^T) = \text{tr}(\underbrace{\Sigma_A^T U_A^T U_B}_{P} \underbrace{\Sigma_B^T}_{Q})$$

we set $w = P^T Q$

$$\rightarrow \text{tr}(A^T B) = \text{tr}(\text{diag}(\sigma_1(A), \dots, \sigma_r(A)) \cdot w \cdot \text{diag}(\sigma_1(B), \dots, \sigma_r(B)))$$

$$= \sum_{i,j} \sigma_i(A) \sigma_j(B) |w_{ij}|^2$$

$$\text{if } \sigma_i = \sigma_i(A), \quad \tau_i = \sigma_i(B)$$

we know that for any boundy stochastic matrix like w , we have:

$$\sum_{i,j} w_{ij} \sigma_i \tau_j \leq \sum_{i=1}^V \sigma_i \tau_i$$

$$\rightarrow \text{tr}(A^T B) \leq \sum_{i=1}^V \sigma_i(B) \sigma_i(A)$$

9)

$$a) \quad t(x) = t(x_0) + (x - x_0)^T \nabla t(x_0) + \frac{1}{2} (x - x_0)^T H (x - x_0)$$

if we set $g = \nabla t(x_0)$, $x = x_0 - \varepsilon g \rightarrow x - x_0 = -\varepsilon g$

$$\rightarrow t(x_0 - \varepsilon g) = t(x_0) + (-\varepsilon g)^T g + \frac{1}{2} (-\varepsilon g)^T H (-\varepsilon g)$$

$$= t(x_0) - \varepsilon g^T g + \frac{1}{2} \varepsilon^2 g^T H g = t(x_0) - \varepsilon \|g\|_2^2 + \frac{1}{2} \varepsilon^2 g^T H g + o(\varepsilon^3)$$

b) as $\varepsilon \rightarrow \infty$, we have $t(x - \varepsilon g) \rightarrow -\infty$

in fact $g^T H g = 0$ and $g^T H g < 0$ would mean flat and negative curvature, so H is not locally PD

hence the approximation becomes unstable.

$$c) \quad \varphi(\varepsilon_0) = t(x_0) - \varepsilon \|g\|^2 + \frac{1}{2} \varepsilon^2 g^T H g$$

$$\rightarrow \frac{d\varphi}{d\varepsilon} = -\|g\|^2 + \varepsilon g^T H g = 0 \rightarrow \varepsilon^* = \frac{\|g\|^2}{g^T H g} \quad \checkmark$$

d) $H = H^T$, so it has orthonormal eigenvectors

$v_1, \dots, v_n$ and eigenvalues $\lambda_1, \dots, \lambda_n$

g can be expanded in eigenbasis:

$$g = \sum_{i=1}^n \alpha_i v_i \quad \text{where } \alpha_i = v_i^T g$$

$$\rightarrow g^T g = \left(\sum \alpha_i v_i \right)^T \sum \alpha_i v_i = \sum \alpha_i^2$$
$$(v_j^T v_i = \delta_{ij})$$

$$H v_i = \lambda_i v_i$$

$$\rightarrow Hg = \sum \alpha_i \lambda_i v_i$$

$$\rightarrow g^T Hg = \sum_{i=1}^n \lambda_i \alpha_i^2$$

$$\rightarrow \varepsilon^* = \frac{\sum_{i=1}^n \alpha_i^2}{\sum_{i=1}^n \lambda_i \alpha_i^2} \quad \checkmark$$

10)

1.

$$\text{tr}(A^T A) = \sum_{j=1}^n (A^T A)_{jj} = \sum_{j=1}^n \left(\sum_{i=1}^m (A^T)_{ji} A_{ij} \right), \quad A_{ji}^T = A_{ij} \rightarrow$$

$$= \sum_{j=1}^n \sum_{i=1}^m A_{ij}^2, \quad A_{ij}^2 = |A_{ij}|^2 \text{ (real matrix)}$$

$$\rightarrow \|A\|_F = \sqrt{\sum_{j=1}^n \sum_{i=1}^m A_{ij}^2}$$

2. SVD: $A = U \Sigma V^T$, $U U^T = I$, $V V^T = I$,

$$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$$

$$\|A\|_F^2 = \text{tr}(A^T A) = \text{tr}((U \Sigma V)^T (U \Sigma V)) = \text{tr}(V^T \Sigma^T U^T U \Sigma V)$$

$$= \text{tr}(V \Sigma^T \Sigma V^T) = \text{tr}(\underbrace{V^T V}_{I} \Sigma^T \Sigma) = \text{tr}(\Sigma^T \Sigma) \quad \overset{I}{\text{tr}}$$

$$\Sigma^T \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_r^2)$$

$$\rightarrow \|A\|_F^2 = \text{tr}(\Sigma^T \Sigma) = \sum_{i=1}^r \sigma_i^2$$

3. Lower Bound: $\sigma_i \geq 0$ for all i

$$\rightarrow \|A\|_F^2 = \sigma_1^2 + \dots + \sigma_r^2 \geq \sigma_{\max}^2 \rightarrow \|A\|_F \geq \sigma_{\max} \quad \checkmark$$

upper bound:

$$\sigma_i \leq \sigma_{\max} \text{ for all } \sigma_i \rightarrow \|A\|_F^2 = \sum_{i=1}^r \sigma_i^2 \leq \sum_{i=1}^r \sigma_{\max}^2 = r \sigma_{\max}^2$$

$$\sqrt{\quad} \rightarrow \|A\|_F \leq \sqrt{r} \sigma_{\max} \checkmark$$

2). if we set b_j , j -th column of B , Ab_j as j -th column

$$\text{of } AB \rightarrow \|AB\|_F^2 = \sum_{j=1}^q \|Ab_j\|_2^2$$

$$\|Ab_j\|_2^2 \leq \sigma_{\max}^2(A) \|b_j\|_2^2$$

$$\rightarrow \|AB\|_F^2 \leq \sum_{i=1}^q \sigma_{\max}^2(A) \|b_j\|_2^2 = \sigma_{\max}^2(A) \sum_{j=1}^q \|b_j\|_2^2$$

$$= \sigma_{\max}^2(A) \|B\|_F^2 \xrightarrow{\sqrt{\quad}} \|AB\|_F \leq \sigma_{\max}(A) \|B\|_F \checkmark$$

from part 3 $\rightarrow$

$$\sigma_{\max}(A) \leq \|A\|_F$$

$$\rightarrow \|AB\|_F \leq \sigma_{\max}(A) \|B\|_F \leq \|A\|_F \|B\|_F \checkmark$$

5.

as discussed in class, this part must consider a symmetric, hence:

$$A = Q \Lambda Q^T, \quad y = Q^T x$$
$$\rightarrow \frac{x^T A x}{x^T x} = \frac{x^T (Q \Lambda Q^T) x}{x^T x} = \frac{y^T \Lambda y}{y^T y} = \frac{\sum_{i=1}^n \lambda_i y_i^2}{\sum_{i=1}^n y_i^2}$$

A is symmetric PSD $\rightarrow$

$$= \frac{\sum_{i=1}^n \sigma_i y_i^2}{\sum_{i=1}^n y_i^2} \leq \frac{\sum_{i=1}^n \sigma_{\max}(A) y_i^2}{\sum_{i=1}^n y_i^2} = \sigma_{\max}(A) \quad \checkmark$$

also for lower hand:

$$\frac{\sum \sigma_i y_i^2}{\sum y_i^2} \geq \frac{\sum \sigma_{\min}(A) y_i^2}{\sum y_i^2} = \sigma_{\min}(A)$$

$$\rightarrow \sigma_{\min}(A) \leq \frac{x^T A x}{x^T x} \leq \sigma_{\max}(A) \quad \checkmark$$

11)

$$1. \quad \|A - X\|_F^2 = \text{tr}((A - X)(A - X)^T) = \|A\|_F^2 + \|X\|_F^2 - 2\text{tr}(A^T X)$$

$$\text{Von Newman: } \text{tr}(A^T X) \leq \sum_{i=1}^{\checkmark} \sigma_i(A) \sigma_i(X)$$

X has at most k non zero singular values $\rightarrow$

$$\text{tr}(A^T X) \leq \sum_{i=1}^k \sigma_i(A) \sigma_i(X)$$

$$\rightarrow \|A - X\|_F^2 \geq \sum_{i=1}^k \sigma_i(A)^2 + \sum_{i=k+1}^r \sigma_i(X)^2 - 2 \sum_{i=1}^k \sigma_i(A) \sigma_i(X)$$

minimal X gives zero squared term, in fact

$$\sigma_i(X) = \sigma_i(A) \text{ so if } X = A_k \quad \|A - A_k\|_F^2 = \sum_{i=k+1}^r \sigma_i(A)^2$$

the min error. $\checkmark$

2. the nullspace of X , denoted as $\ker(X)$ has at least $n-k$ dimensions (by rank-nullity theorem), now we consider subspace $\mathcal{V}_{k+1}$ spanned by the first $k+1$ columns of $\mathcal{V}$, having dimension of $k+1$.

$$\dim(\ker(X)) + \dim(\mathcal{V}_{k+1}) \geq (n-k) + (k+1) > n$$

$$\text{so } \exists z \mid z \in \ker(X) \cap \mathcal{V}_{k+1}, \|z\|_2 = 1$$

$$z \in \mathcal{V}_{k+1} \rightarrow z = \sum_{i=1}^{k+1} c_i v_i, \sum c_i^2 = 1$$

$$z \in \ker(X) \rightarrow Xz = 0, (A-X)z = Az$$

$$Az = A \left(\sum_{i=1}^{k+1} c_i v_i \right) = \sum_{i=1}^{k+1} (c_i A v_i) = \sum_{i=1}^k c_i \sigma_i v_i$$

$$\|(A-X)z\|_2^2 = \|Az\|_2^2 = \sum_{i=1}^{k+1} c_i^2 \sigma_i^2$$

$$\sigma_1 > \sigma_2 \dots > \sigma_{k+1} \rightarrow \sum_{i=1}^{k+1} c_i^2 \sigma_i^2 \geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} c_i^2 = \sigma_{k+1}^2$$

$$\rightarrow \|(A-X)z\|_2 \geq \sigma_{k+1} \quad (I)$$

$$\|A-X\|_2 = \max_{\|y\|_2=1} \|(A-X)y\|_2, \|z\|_2=1$$

$$\rightarrow \|A-X\|_2 \geq \sigma_{k+1}, \text{ if } X = A_k$$

$$\rightarrow \|A - A_k\|_2 = \sigma_{k+1} \quad \checkmark$$